Globally-centered autocovariances in MCMC*

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MCMC Setup

Let F be an **intractable** absolutely continuous target distribution on $\mathcal{X} \subset \mathbb{R}^d$ with associated Lebesgue density function $f : \mathcal{X} \to \mathbb{R}$.

Interested in features of F like

$$\mu_g := \mathbb{E}_X[g(X)] = \int_{\mathcal{X}} g(x)f(x)dx$$

$g : \mathcal{X} \to \mathbb{R}^p$ is an F-integrable function.

Markov chain Monte Carlo algorithms provide tools for constructing ergodic Markov chains with stationary distribution F.

We have Markov kernel $P : \mathscr{X} \times \mathscr{B}(\mathscr{X}) \to [0,1]$ such that for any initial distribution λ

$$\|\lambda P^n - F\|_{\mathrm{TV}} \to 0$$

Obtain the ergodic Markov chain $X_1, \ldots, X_n$ using the above kernel. Let $Y_i := g(X_i)$. Using Birkhoff's ergodic theorem,

$$\bar{Y} := \frac{1}{n}\sum_{i=1}^{n} Y_i \to \mu_g \quad \text{with probability } 1 \text{ as } n \to \infty\,.$$

The asymptotic distribution of $\bar{Y}$ is given by Markov chain CLT

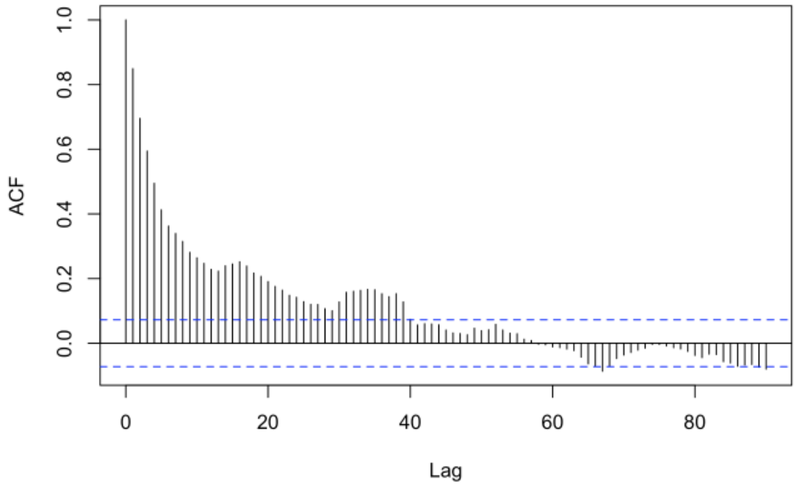
$$\sqrt{n}(\bar{Y} - \mu_g) \xrightarrow{d} \mathscr{N}(0,\Sigma)$$

where $\Sigma = \sum_{k=-\infty}^{\infty} \mathrm{Cov}_F(Y_1, Y_{1+k}) =: \sum_{k=-\infty}^{\infty} \Upsilon(k)\,.$

MCMC performance assessment

Autocorrelation Plots

Plot the evolution of lag-correlations. Higher decay is preferable.



Estimated Asymptotic Variance of Monte Carlo estimator

The covariance matrix

$$\Sigma = \sum_{k=-\infty}^{\infty} \mathrm{Cov}_F(Y_1, Y_{1+k}) =: \sum_{k=-\infty}^{\infty} \Upsilon(k)$$

can be estimated using methods:

- **Spectral variance estimator**
- Batch means estimator

Effective Sample Size

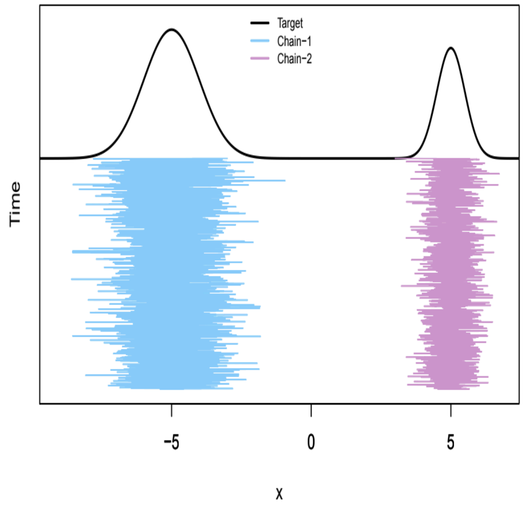
Number of i.i.d. samples that would yield same variability in $\bar{Y}$ as the correlated sample obtained from MCMC.

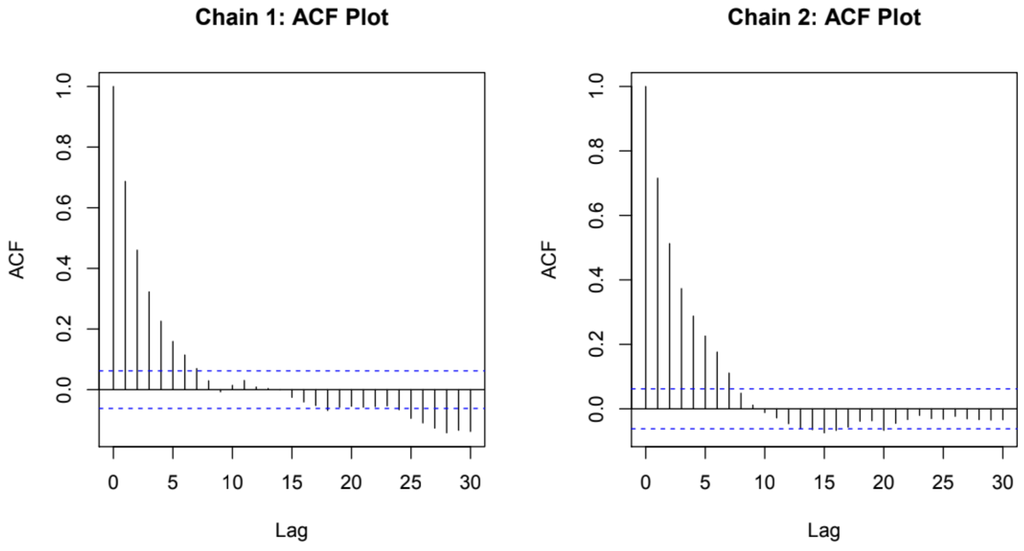
$$\mathrm{ESS} = n \left(\frac{|\Upsilon(0)|}{|\Sigma|} \right)^{1/p}$$

Problem Demonstration

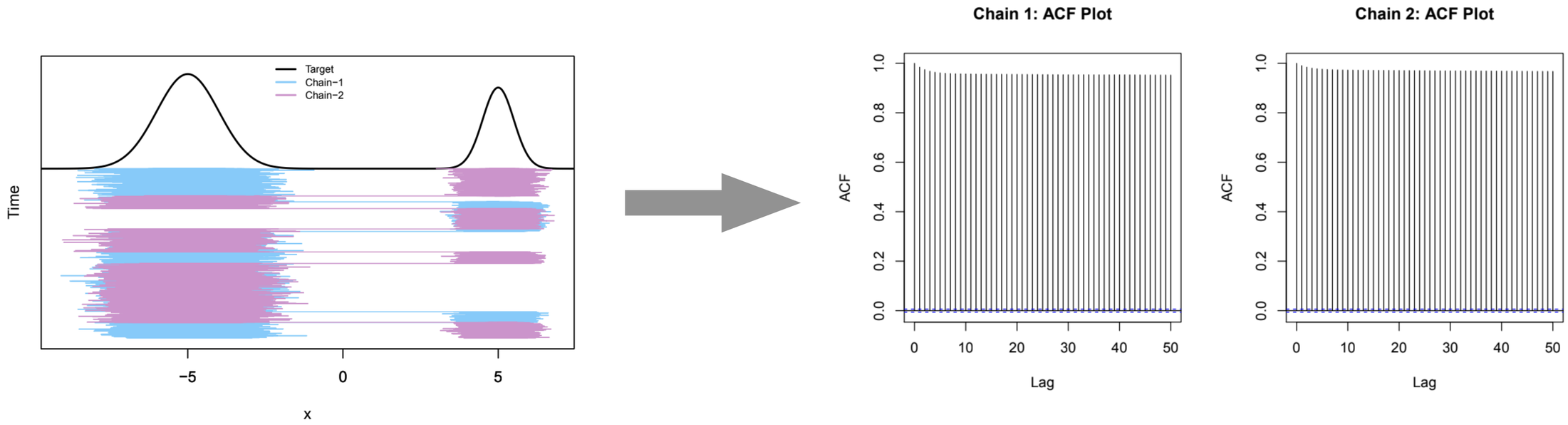
Gaussian Mixture

Consider a mixture of Gaussian distributions with two well-separated modes. We run two stationary Metropolis-Hastings implementations for $n = 10^3$.





Gaussian Mixture for longer run



Running the Markov chains for 10^5 steps reveals that the Markov chains are actually very correlated.

The ACF plot for both chains critically underestimates the lag-correlation for $n = 10^3$.

What's wrong?

For the Markov chain $X_1, \ldots, X_n$, the sample auto covariance estimator is

$$\hat{\Gamma}(k) = \frac{1}{n} \sum_{i=1}^{n-k} (X_t - \bar{X})(X_{t+k} - \bar{X})^\top$$

For a single chain, the mean $\bar{X}$ is the local center of the chain which is centered at the local mode. Variability around this mode is far lesser than variability around actual mean.

For slow-mixing Markov chains, the empirical auto covariance can highly underestimate the truth!

Impact of slow-mixing Markov chains

For slow mixing Markov chains:

- The autocorrelation function (ACF) plots can be misleading if the chain is not run long enough.
- Since asymptotic variance Σ is sum of lag-covariances, it is underestimated too.
- The ESS is misleadingly inflated causing early termination of MCMC procedure.

Parallel Markov chains

Suppose one wants to run parallel Markov chains with dispersed starting points to more accurately capture the target distribution.

Let $\{X_{s,t}\}_{t\geq 1}$ be the sth Markov chain for $s = 1,\ldots,m$ and let $Y_{s,t} = g(X_{s,t})$.

Let $\bar{Y}_s$ be the sample mean of sth chain. They can be vastly different for slow-mixing chains!

The global mean is $\bar{\bar{Y}} = \dfrac{1}{m}\displaystyle\sum_{s=1}^{m} \bar{Y}_s$.

The autocovariance function (ACvF) for sth chain is $\hat{\Upsilon}_s(k) = \dfrac{1}{n}\displaystyle\sum_{t=1}^{n-k} (Y_{s,t} - \bar{Y}_s)(Y_{s,t+k} - \bar{Y}_s)^\top$.

Globally-centered estimators

1. Globally-centered autocovariance function (G-ACvF)
2. Globally-centered spectral variance (G-SV) estimator
3. Globally-centered effective sample size (G-ESS)

Globally-centered auto covariance estimator

There is no unified approach to constructing estimators of $\Gamma(k)$ for parallel-chain implementations.

We propose globally-centered autocovariance estimator

$$\hat{\Upsilon}_{G,s}(k) = \frac{1}{n}\sum_{t=1}^{n-k}(Y_{s,t} - \bar{\bar{Y}})(Y_{s,t+k} - \bar{\bar{Y}})^{\top}$$

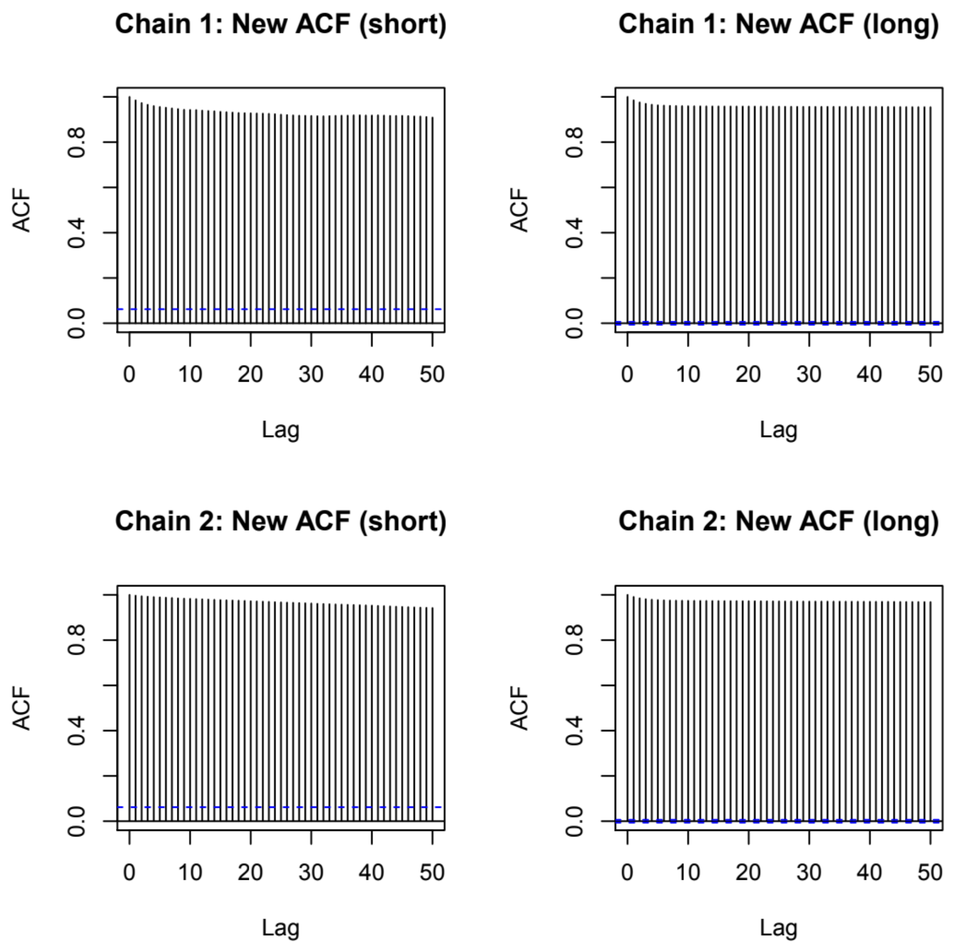
and the overall globally-centered autocovariance estimator is

$$\hat{\Upsilon}_{G}(k) = \frac{1}{m}\sum_{s=1}^{m}\hat{\Upsilon}_{G,s}(k)\,.$$

In the presence of positive autocorrelation, both $\hat{\Upsilon}_{s}(k)$ and $\hat{\Upsilon}_{G,s}$ are under biased and asymptotically unbiased. G-ACvF estimator leads to reduction in bias.

Coming back to Gaussian mixture example...

$n = 1000$



$n = 100{,}000$

Globally-centered ACF plots

Globally-centered estimators

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Globally-centered MCMC asymptotic variance

Markov chain CLT tells us that for each chain

$$\sqrt{n}(\bar{Y}_s - \mu_g) \xrightarrow{d} \mathcal{N}(0,\Sigma)\,.$$

Since each Markov chain is run independently,

$$\sqrt{mn}(\bar{\bar{Y}} - \mu_g) \xrightarrow{d} \mathcal{N}(0,\Sigma)\,.$$

where $\Sigma = \sum_{k=-\infty}^{\infty} \Upsilon(k)\,.$

G-ACvF estimator for $\Upsilon(k)$ could lead to improvement in estimation of limiting variance of $\bar{\bar{Y}}$. We use spectral variance estimator.

Spectral Variance Estimator

Spectral variance estimator is a weighted and truncated sum of estimated ACvFs.

Lag Window

Truncation Point

- For $c \geq 1$, $w : \mathbb{R} \to [-c, c]$ be the lag window function that is continuous on all but a finite number of points, is a bounded and even function with $w(0) = 1$, $\int w(x)^2 dx < \infty$, and $\int |w(x)| dx < \infty$.
- We use the Bartlett lag window — $w(x) = 1 - |x|$ for $|x| < 1$ and 0 otherwise.
- Truncation point $b_n \in \mathbb{N}$ determines the number of lag covariances used in estimating Σ.
- SV estimator for sth chain is $\hat{\Sigma}_s := \sum_{-b_n+1}^{b_n-1} w\left(\frac{k}{b_n}\right) \hat{\Upsilon}_s(k)$.

Globally-centered spectral variance estimator

A natural estimator for Σ using m chains is the average SV (ASV) estimator $\hat{\Sigma}_A := m^{-1} \sum_{s=1}^{m} \hat{\Sigma}_s$.

Large sample properties of $\hat{\Sigma}_s$ extend directly to $\hat{\Sigma}_A$.

We define G-SV estimator as the weighted and truncated sum of G-ACvFs.

$$\hat{\Sigma}_G = \sum_{k=-b_n+1}^{b_n-1} w\left(\frac{k}{b_n}\right) \hat{\Upsilon}_G(k)\,.$$

Theoretical results for G-SV estimator

Strong Consistency: Let $\mathbb{E}[\|Y_{1,1}\|^{2+\delta}] < \infty$ for $\delta > 0$, P be polynomially ergodic with order $\zeta > (2+\epsilon)\big/(1+2/\delta)$. If $\hat{\Sigma}_s \xrightarrow{a.s.} \Sigma$ for all s and $n^{-1}b_n \log\log n \to 0$, then $\hat{\Sigma}_G \xrightarrow{a.s.} \Sigma$.

Large Sample Bias: The asymptotic result for bias incurred in estimating Σ using $\hat{\Sigma}_G$ is the same as that using $\hat{\Sigma}_A$ provided $\dfrac{b_n^{q+1}}{n} \to 0$ where q is such that $\lim\limits_{x\to 0} \dfrac{1-w(x)}{|x|^q} \to k_q < \infty$.

Large Sample Variance: The asymptotic result for variance of $\hat{\Sigma}_G$ is also same as $\hat{\Sigma}_A$.

Globally-centered estimators

1. Globally-centered autocovariance function (G-ACvF)
2. Globally-centered spectral variance (G-SV) estimator
3. Globally-centered effective sample size (G-ESS)

Globally-centered ESS

ESS are the number of independent and identically distributed samples that would yield same Monte Carlo variability in $\bar{\bar{Y}}$ as the correlated sample.

$$\text{ESS} = mn \left(\frac{|\Upsilon(0)|}{|\Sigma|} \right)^{1/p}$$

To obtain more reliable estimates of the ESS, we use the G-SV estimator for Σ:

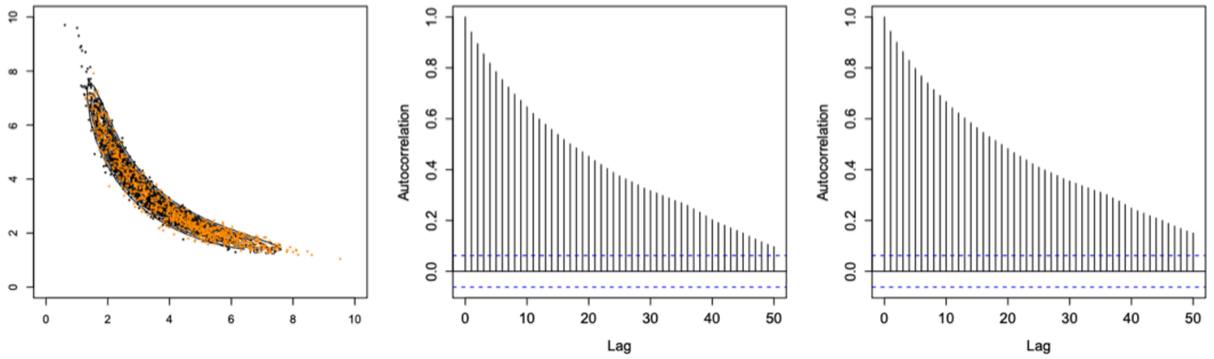
$$\widehat{\text{ESS}}_G = mn \left(\frac{|\hat{\Upsilon}(0)|}{|\hat{\Sigma}_G|} \right)^{1/p}$$

ESS is used in determining when to stop an MCMC simulation by terminating the process when estimated ESS is greater than a pre-specified lower bound.

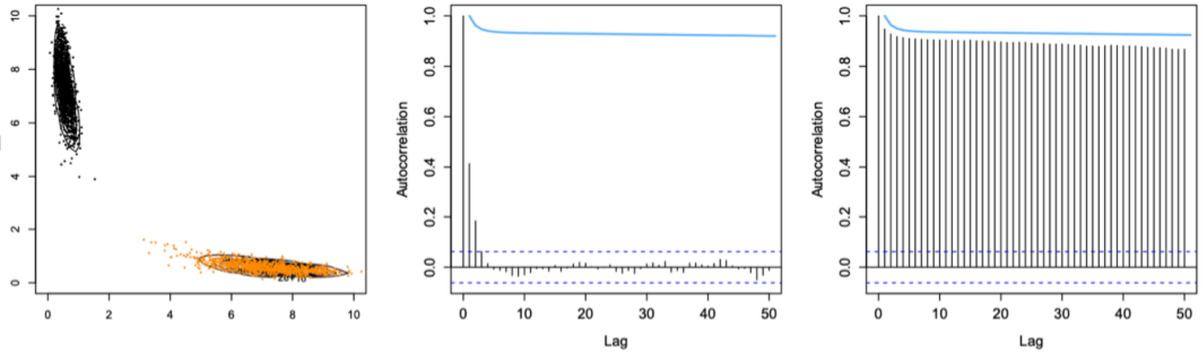
Example: Boomerang Distribution

For $A \geq 0$ and $B, C \in \mathbb{R}$, the target density is $f(x, y) \propto \exp\left(-\frac{1}{2}\left[Ax^2y^2 + x^2 + y^2 - 2Bxy - 2Cx - 2Cy\right]\right)$.

Setting 1:



Setting 2:



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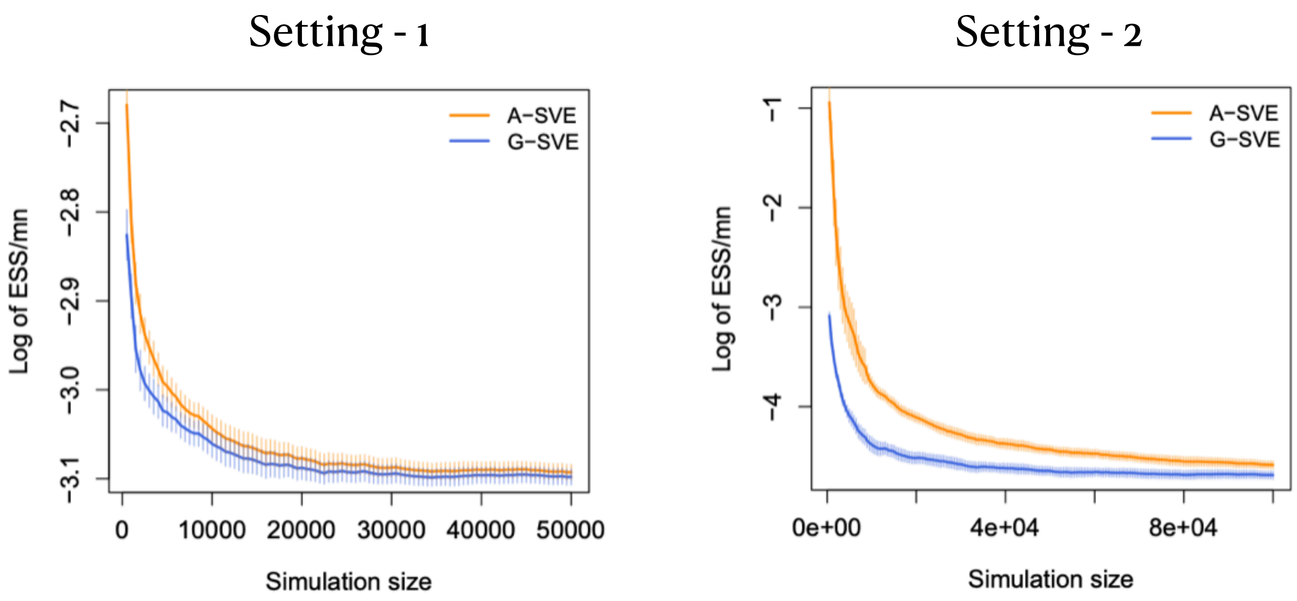
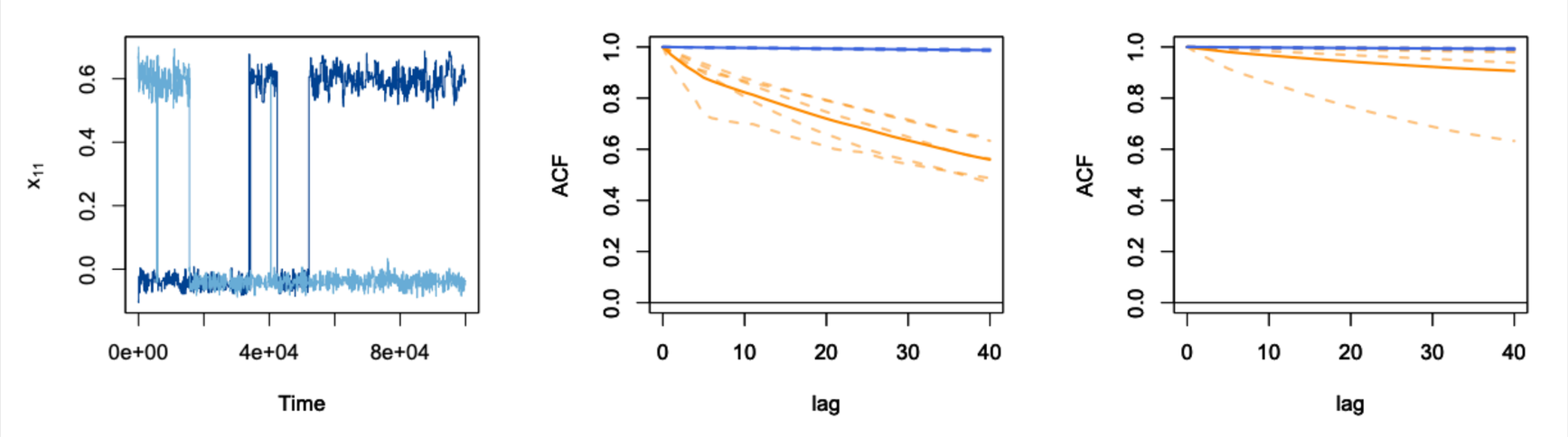


Running plot for $\log(\widehat{\mathrm{ESS}_A})/mn$ and $\log(\widehat{\mathrm{ESS}_G})/mn$ with $m = 5$ for setting 1 (left) and setting-2 (right).

Example: Sensor Network Localization

The goal is to identify unknown sensor locations using noisy distance data. Let $x_i = (x_{i1}, x_{i2})$ denote the coordinates for the position of the ith sensor. A Metropolis-within-Gibbs type sampler is implemented. We run $m = 5$ parallel Markov chains with randomized starting points.



Trace Plot got two Markov chains

ACF plots for $n = 5 \times 10^3$

ACF plots for $n = 5 \times 10^4$

Trace and ACF Plots for x_{11}

Conclusion

- Even parallel Markov chains can give deceiving convergence diagnostics for slow-mixing Markov chains when naive averaging is used.
- Information from parallel Markov chains can be pooled using the simple global-centering trick.
- The estimators obtained are “asymptotically similar” and “at least as good” as estimators obtained through naive averaging in finite samples case.
- The ACF plots are made using the `R` package `multichainACF` available on GitHub.

Thank you!
Questions?